DAANVIA

Built-in navigation for your motorcycle display

Revision 3: offline navigation, parking pins and adaptive TFT layouts
Updated 19 October 2026 | Cebu first, Philippines

Decision

Move from the proven connection prototype to a navigation MVP. Preserve the working Wi-Fi Direct / H.264 transport; keep the scanner and logs under setup and support. The baseline is open-source and on-device. The owner prefers more engineering over recurring per-trip/provider fees. Commercial navigation SDKs are alternatives only, not a prerequisite.

What is proven

Owner tests reported visible output from an independent app on one ATR 160 dashboard and one OPPO Android 16 phone. The latest reported pattern and mirror sessions ran to their 45-second and 90-second test limits. This establishes transport feasibility for that combination; it does not establish long-ride reliability or support for every firmware revision.

What we are building

Connect bike > choose destination > compare routes > start ride. The phone owns GPS, routing, voice and rendering. A dedicated navigation scene goes to the bike. Near the destination, arrival mode passively shows parking pins with motorcycle prices directly on the map before arrival. No parking footer, details panel or selection controls; missing prices have no price label. Final navigation must work without screen-capturing another app.

Working identity

Daanvia combines the Filipino word for road with a route-oriented name. Three original visual directions and logo concepts accompany this brief. The name is provisional: a web search is not trademark clearance, domain ownership or app-store reservation. Do IPOPHL and relevant international trademark searches before spending on launch branding. Generated artwork remains a concept until refined and checked for distinctiveness.

Recommendation

Adopt OpenStreetMap-derived regional data and MapLibre Native. Evaluate Valhalla as the first on-device routing candidate, with GraphHopper as a comparison; choose through an Android integration spike, not a server-only demo. Include offline search and parking prices near the destination in the MVP. Internet services and AI are optional enhancements, never required for a basic ride.

Reading this brief

Published facts carry source IDs. Architecture, budgets, quality thresholds and rollout steps are recommendations. Prices exclude tax and currency conversion. None of the vendor contacts, group posts, purchases or paid subscriptions described here has been sent or created.

Offline-first architecture: the baseline decision

The baseline must calculate routes, reroute, search downloaded places, announce turns and draw the bike map without mobile data. The phone GPS supplies position; no paid cloud service is called per GPS fix, map frame or trip. More engineering and local storage are accepted tradeoffs.

Component	Proposed implementation	Verification gate
Geographic data	OSM-derived Cebu extract with required attribution and ODbL review.	Check motorcycle restrictions, one-way rules, routing access and parking entrances.
Map display	MapLibre Native with a licensed/self-produced vector map package, style, glyphs, fonts and sprites.	Offline map rendering at required zooms; adaptive safe viewport, custom screen layouts and screen-off operation.
Routing	Valhalla first candidate via Android NDK/JNI; compare GraphHopper if necessary.	Demonstrate a real local route and missed-turn reroute with networking disabled. Native build, ABI, memory, speed and licensing must pass.
Search / parking	SQLite spatial index / FTS with regional POIs, parking entrances and tariff records.	Offline search and access-aware parking pins; unknown fields remain unknown internally and missing prices have no TFT label.
Guidance	Local progress/map matching, maneuver timing, reroute debounce and installed offline Android TTS voice.	Noisy GPS and complex junction tests; do not repeat or skip maneuvers after reconnect.
Display link	Reuse tested Java Wi-Fi Direct/PXC/H.264 transport; Kotlin for new layers.	Independent rendering plus saved safe-area/layout profiles; reconnect and thermal tests.
Updates	Signed/versioned regional downloads from our own hosting or a provider permitting offline use.	Checksum, schema validation, resumable download, atomic install and rollback.

A region package is more than map tiles

Package map graphics, routing graph, POI/search indexes and parking/rate data with compatible versions and separate layers. Map tiles alone cannot reroute. MBTiles/PMTiles may store map graphics, subject to the selected Android reader; they are not the routing graph. Build and test an actual supported local file reader. [S17,S26,S27]

Commercial providers remain optional

Google/TomTom/HERE/Mapbox are comparison or optional licensed enhancement paths. Do not combine their route/POI/traffic content with an OSM map without checking the exact agreement. No provider fee is required for the proposed offline baseline. Public OSM tile servers prohibit offline bulk prefetch; use lawful packages or our own generation/hosting. [S6-S16,S32,S33]

Libraries and reference repositories

Use native Android Kotlin for new app layers; retain the tested Java transport through normal interoperability. Latest is a research input, not permission to upgrade every dependency at once. Lock a compatible tested set in Gradle and maintain a dependency/license inventory.

Layer	Candidate / verified snapshot	Use and caution
Phone UI	Kotlin 2.4.21; Compose BOM 2026.09.00; Material 3 [S20,S21]	Navigation planning and accessible settings. Confirm Android Gradle Plugin / Kotlin / SDK compatibility at project bootstrap.
QR setup	CameraX stable 1.6.2; ZXing 3.5.4 [S22,S41]	Replace legacy camera integration when migrating setup; local QR decoding, strict parser and credential redaction. Prototype currently bundles ZXing 3.5.3.
Map rendering	MapLibre Native Android 13.7.1, BSD-2-Clause [S17]	Flexible map styling. Do not force this version if navigation dependencies require a different tested version.
Navigation logic	MapLibre Navigation release 5.0.0; MIT [S18,S19]	GitHub release and Maven core metadata disagree: latest published core is 5.0.0-pre14. Verify artifacts/API maturity before treating the release as a drop-in stable dependency. [S43]
Networking	OkHttp 5.5.0, Apache-2.0 [S42]	HTTP API calls, timeouts and cancellation. Keep proprietary dashboard framing isolated from HTTP code.
Android services	Fused Location or platform LocationManager; MediaCodec/EGL; TextToSpeech	Single location feed, dedicated hardware video encoding and voice. These platform choices do not require cloud AI per GPS update.
Persistence / support	Room; DataStore; WorkManager; structured redacted logs	Saved destinations, preferences and deferred maintenance. WorkManager is not the active navigation/GPS loop. Choose current compatible stable releases at bootstrap.

Open-source repos to read, with boundaries

MOTO-HUB CORE (AGPL-3.0): pairing, network recovery, handlebar events and connection UX. Its full navigation ADV-SOLO product is closed source, so it is not an open navigation codebase to copy. **ridedaemon-lib** (GPL-3.0): session, heartbeat and media transport reference. **open-cflink**: protocol documentation and sample flow; no top-level license detected, so do not copy its code without permission. [S23-S25]

Valhalla 3.9.1 (MIT): route costing/server design. **GraphHopper** 11.1 (Apache-2.0): graph routing and custom models. **Organic Maps** (main license Apache-2.0): offline-first navigation, search and UX reference; it is a substantial app, not a small drop-in SDK. Review all bundled component and data licenses. Repo activity/releases checked through official GitHub API. [S26-S28]

GPL/AGPL software can be sold, but distributing a derivative app can require corresponding source and license compliance. Keeping a module separate does not automatically avoid those obligations. For a closed-source product, audit provenance or seek an alternative license before reuse.

A calm, usable navigation experience

Primary flow: one-time QR setup, destination search, route preview, Start ride, automatic dashboard connection/recovery, and an obvious End ride action. No mandatory account for local-only navigation unless a selected provider or commercial feature requires one.

Phone while parked

Use Material 3 conventions, familiar back behavior and one dominant action per screen. Show destination entrance, travel time, distance and selected motorcycle profile before starting. Choose planned parking duration, walking limit and price preference before departure; show a route preview and expected parking costs. Arrival parking is a core feature, not a hidden optional screen. Request camera, nearby-device and precise-location permissions only when the relevant feature needs them.

Bike while riding

Use 800x480 as the tested ATR transport canvas, not a universal physical or unobstructed display size. Layout must use calibrated visible bounds and reserved areas. Prioritize turn arrow, distance to maneuver, street name and a simplified 2D map. ETA and distance are secondary. Arrival mode adds up to three stable parking pins with known prices attached directly to each pin; no parking footer or selection panel. Hide developer controls and diagnostics. Preserve any original vehicle indicators or warning areas that remain outside the projection viewport; do not imitate instrument readings or assume the display supports touch.

Accessibility and legibility

Phone controls should provide at least 48x48 dp touch targets; use larger practical targets where appropriate. Target WCAG AA text contrast, support font scaling and TalkBack on the phone, and never convey route status by color alone. Driver-oriented guidance favors short text and glanceable content; it is a design reference, not Android Auto certification of our custom protocol. Test daylight glare, darkness and vibration on the actual bike. [S30,S31]

Honest states

Condition	What the rider sees
Finding GPS	Finding your location; Start ride waits for a usable position.
Poor / stale GPS	Location signal weak; avoid showing an advancing position as if confirmed.
No data connection	Traffic unavailable; show cached guidance only if supported, and identify stale information.
Bike disconnected	Voice continues if usable; phone indicates reconnecting with a clear stop option.
Off route	Recalculating; do not oscillate instructions after one noisy GPS point.
Parking arrival	Known applicable rates appear on pins with their basis. Unknown or unreliable rates have no price label; do not claim live occupancy.

Reference reviewed

The Mobbin Google Maps route-preview screen shows map-first planning, time/distance, a route warning and a clear Start action. Use those hierarchy principles; retain original branding and layout. The reviewed reference is linked in [S40]. The generated maps in our concepts are illustrative and must be replaced by licensed geographic data.

Fit my TFT: adaptive layouts and saved profiles

Owner requirement: dashboards differ in aspect ratio, dimensions and native overlays. The ATR currently shows a footer during mirroring that can hide content and make a portrait phone screen appear cropped or distorted. Do not assume the full video canvas is visible or that the native footer can be removed.

A navigation scene designed for the bike

Render a dedicated navigation view independently from the phone UI. Negotiate supported video dimensions where the protocol permits, then reflow components into the calibrated safe viewport. Preserve geometry and font proportions; never non-uniformly stretch the UI to fill a different shape. Scale/letterbox when needed. Cropping may apply to map framing only, never hide turn instructions, prices, attribution or status text.

Phone setup: Fit my TFT

While parked, select a saved bike/profile or start a generic profile. Read reported transport capabilities and propose a canvas. Send a calibration screen with visible corner targets, grid, round circle and readable size samples. The phone shows a preview and draggable safe-area edges plus numeric/nudge controls for top, bottom, left and right. Reserve a footer or other overlay as an exclusion area. Adjust zoom/offset only within validated bounds, then save or reset. The bike view updates during calibration so the rider can compare it directly.

Reported size versus visible size

Show encoder canvas resolution separately from usable viewport and reserved insets. Manual visible-area calibration remains necessary if the firmware does not report native overlays. Do not label a phone preview as a live screenshot of the TFT unless we actually receive a display capture. A saved profile is a user override, not automatic evidence of firmware behavior.

Editable setting	Behavior
Safe bounds / footer	Top/bottom/left/right insets and optional exclusion rectangles, saved in normalized coordinates plus the source canvas size; validate a minimum useful area.
Layout selection	Map focus, large-turn focus and arrival parking variants. Responsive rules move/shrink secondary panels, reduce parking candidates and abbreviate labels before reducing core readability.
Readability / appearance	Bounded text scale, day/night palette, map label density and contrast. Keep units and distinctions for parking prices legible; do not allow hiding critical guidance.
Profile identity	Store per saved dashboard with model/firmware metadata where known, orientation, supported canvas, fit mode and layout preferences. Reconfirm if capability/aspect changes; do not reuse stale pixel offsets blindly.
Safety / recovery	Editor available only while parked; Cancel restores previous configuration, Reset uses safe defaults, changes persist after reconnect/restart. Roll back invalid profiles.

Acceptance criteria

Verify known ATR footer reserved area on physical hardware; circles remain round and text undistorted; all required labels fit. Simulate wide, compact and differently shaped viewports, overlapping exclusions, font scaling and portrait phone rotation. No parking label or maneuver may land under an overlay. If the usable area is too small, show a compact turn-first fallback and explain the limit in setup rather than silently clipping content.

Reliability and battery are core features

Architecture

Keep five clear modules: bike connection, navigation engine, location/voice service, display renderer/encoder, and phone UI. Keep local routing and regional packages behind clear interfaces; optional cloud enhancements must not break offline navigation or violate data licenses. Model session state explicitly: connecting, connected, navigating, reconnecting, stopped. Own callbacks and sockets within a single session; make start/stop idempotent.

Screen-off navigation is a development gate

Render our navigation scene into a dedicated encoder surface or supported offscreen map pipeline. Avoid dependence on MediaProjection, a foreground phone Activity, screenshots of another app, or a per-frame network map-image request. The chosen map SDK must both permit and technically support this rendering path. Prototype it early; it is not yet proven by the screen-mirroring tests.

Battery strategy

Use one navigation location feed with suitable accuracy/update interval while moving and less activity while stationary. Keep interpolation and display animation separate from GPS frequency. Prefer simple 2D rendering and hardware H.264 at 800x480; evaluate roughly 10-15 fps rather than assuming more frames improve navigation. Cache only as licensed. Stop location, encoding and network work at trip end. Android documents frequent foreground navigation updates and reducing unnecessary location work. [S29]

Proposed release gates - targets, not measured results

Test	Acceptance evidence
1-2 hour navigation	Stable output with phone locked, no unhandled crash, no leaked connection, no sustained thermal degradation.
Interruptions	Incoming call, app switch, screen off/on, GPS loss, mobile-data loss, bike power cycle and Wi-Fi coexistence. Recover with clear state; never duplicate sessions.
Offline correctness	Networking disabled: local search, route calculation, missed-turn reroute, voice, parking rates and GPS progress. Review 20 local routes, restricted roads and entrances.
Protocol resilience	Bounds on message/frame sizes, unknown commands, truncated traffic, heartbeat timing, port reuse, reconnect and codec keyframe recovery. Audit acknowledgments before shipping.
Privacy and updates	Logs redact QR secrets, identities and exact location by default. Consent before support export. Verify signed update/rollback behavior.
Battery comparison	Same route, phone, brightness, temperature and network. Compare battery energy/hour, device temperature, CPU/GPU time, data and disconnects. No advertised savings until measured.

Keep the useful diagnostic report as a support tool. Add tests for real failure modes and navigation behavior, not an endless sequence of new diagnostic screens. Test on multiple phones and ATR firmware revisions before widening compatibility claims.

Parking before arrival: required MVP behavior

The rider should see nearby motorcycle parking on the TFT before reaching the destination, including comparable prices where reliable data exists. This is a primary product requirement. It must work from the downloaded regional package when offline.

Journey and trigger

During route preview, select planned parking duration (for example two hours), maximum walking distance and preference: cheapest known price or shortest walk. Prepare candidates then. Proposed arrival trigger: remaining route distance at most 1 km OR route ETA at most 3 minutes. Tune with rides; use debounce and hysteresis so GPS noise does not repeatedly switch modes. Preload candidates earlier, not only when connectivity may already be lost.

TFT information hierarchy

Keep turn arrow, maneuver distance and next street visible. Show destination distinctly from parking. Display one to three non-overlapping P parking pins according to the active display profile. Attach a short known price and its basis directly to the pin, for example PHP40/2h. A lot with no reliable price keeps its parking icon with no price text, question mark or unknown badge. No parking footer, comparison strip, sidebar, selected-lot highlight, selection controls or automatic carousel. Parking is passive information on the TFT. Hide or reduce parking detail near complex/imminent maneuvers; retain the driving instruction.

Parking state	Display rule
Current complete tariff	Calculate estimated total for selected duration/time and motorcycle category; show basis on the pin, e.g. PHP40 / 2h. Phone details while parked retain source and verification date.
Hourly / entry / first-period rate	Do not compare unlike prices as if totals. Show PHP20/hour, PHP30/entry, or first-period wording if a full total cannot be calculated.
Old, partial or conflicting tariff	Omit the price label for missing, stale beyond policy, conflicting or otherwise unreliable rates. Retain the P icon; never show unknown text, question marks or PHP0 as a replacement.
Free parking	Only label free when an explicit, applicable and sufficiently fresh record supports it. Missing fee tags do not establish free parking.
Occupancy / opening hours	No occupancy footer or parking status strip. Do not imply spaces are free without a current occupancy feed. Apply known access/hours to candidate filtering; capacity is not a live space count.
Route choice	Pins cannot be selected on the TFT and never change the destination automatically. Any optional explicit parking destination is set on the phone while parked; no TFT interaction is required.

Passive pins and the final meters

Route to the motorcycle entrance, not a building centroid or inaccessible back wall. Account for one-way approach and public/customer/private access. If a parking destination was explicitly set on the phone before riding, retain the original destination for a walking leg after parking. Otherwise keep guiding to the original destination while displaying nearby pins. With no known suitable lots, show no parking pins; never invent one. Required attribution may remain in the map safe area, but not as a parking footer.

Parking data: ownership, pricing and offline storage

Accurate fees and entrances are the main data challenge. Start with 10-20 surveyed/permissioned Cebu lots around a few destinations rather than claiming complete city coverage. OSM includes parking, motorcycle parking, entrance, access, fee and charge tags, but optional tags do not guarantee complete or current local tariffs. [S44-S47]

Regional record design

Store stable lot ID, coordinates and entrance geometry; vehicle eligibility; access/public/customer restrictions; opening hours/time zone; tariff currency and structured rules; source and permission/license; verified-at and effective dates; freshness/confidence state; and any correction status. Keep live occupancy in a separate expiring layer. Retain raw source evidence privately only where permitted; do not publish reporter identities.

Tariff calculation is deterministic

Support first period plus subsequent increments, flat entry, daily caps, weekday/weekend and time bands, motorcycle-specific rates and explicit grace periods. Calculate the planned stay against the actual schedule and tariff version. If a reliable listed basis is usable, show that basis without inventing a total. Otherwise omit the TFT price label; retain the data-quality reason in phone details. A changed arrival time can change price; recalculate locally.

Initial data workflow

Use operator-published/permissioned information, direct verification and optional rider submissions. Photograph a rate board only with appropriate permission and privacy handling; record the original source, applicable vehicle and date. Moderation verifies and versions corrections. Do not scrape Google Maps/Places into our permanent parking database.

Offline delivery and freshness

Ship parking POIs and rates as a versioned SQLite layer alongside maps/search/graph, allowing small independent updates. Before departure, offer an update when available; retain the last good package if download fails. A proposed initial policy marks tariffs older than 30 days stale in phone details and suppresses their TFT amount until reverified; tune per operator reliability. The TFT retains the parking icon without a missing/stale-price badge. These thresholds are product defaults, not guarantees that a newer price is correct.

Data quality tests

Test motorcycles rejected, entrance across a one-way street, customer-only lots, overnight stays, midnight/weekend boundaries, missing fees, stale tariffs, conflicting reports, no occupancy feed and interrupted updates. Verify pin placement and price labels at calibrated ATR output and smaller/wider profile simulations, in sunlight, without covering guidance or the native footer. Rate tests must include fixture examples with known totals and boundary times.

Useful AI without a recurring dependency

Use	Value and boundary
Parking preference parsing	Interpret a parked request such as cheap motorcycle parking within five minutes walk for two hours. Convert it to explicit constraints for local search/ranking; deterministic controls remain available without AI.
Rate-board extraction	Optional on-device OCR extracts text; optional AI proposes structured tariff fields. A reviewer confirms vehicle category, time bands, amount and date before publishing. This is an authoring aid, not a source of truth.
Explain a parking choice	Summarize verified fields: estimated PHP40 for two hours and 220m walk versus PHP60 and 80m. A simple template already works offline; an LLM is not necessary.
Closure notices	Optional AI extracts locations/times from permitted official notices for review. An event or festival mention alone must not close a road in routing.
Trip planning	Optional online AI can propose stops or interpret natural language when parked. Display provenance and let local routing validate access and the route. Core navigation survives AI failure.

What should not use an LLM

GPS tracking, graph search, route progress, safety restrictions, price arithmetic, tariff validity, pin placement and parking ranking should use ordinary tested code. Cheapest/nearest ranking is deterministic. AI cannot infer empty spaces, current fees or legal road access from the map.

Cost policy

Baseline app works with no AI account, no mandatory AI server and no paid inference calls. Prefer local OCR and templates. Cloud AI, live traffic and occupancy feeds are individually opt-in, with explicit budgets and a working off switch. A local LLM is a later experiment only if its memory, model size, latency and battery cost are justified.

Interaction while riding

The TFT shows the result, not an AI conversation. At most a brief spoken arrival summary, configurable before riding. Do not require riders to read long explanations, confirm a complex chat response or enter tariff data while moving.

Cost model: no mandatory per-trip services

Owner decision: accept more engineering in exchange for reducing ongoing costs. The selected baseline has no paid navigation SDK, no billed cloud route request and no mandatory cloud AI call. Software license fees of zero do not mean packaging, data verification and distribution cost nothing.

Cost item	Baseline design	Remaining expense
GPS / guidance	Phone sensors, local logic and installed offline TTS voice.	Battery, engineering and testing; no per-fix cloud mapping fee.
Map display / routing	MapLibre plus permissively licensed engine and OSM-derived data.	Map/graph build machine time, integration, ODbL/attribution compliance and map updates.
Regional downloads	Versioned map, routing, search and parking packages.	Storage and download bandwidth, even if every ride afterward is offline.
Parking prices	Curated local database with correction workflow.	Surveying, operator verification, moderation and updates; comprehensive accurate rates are not automatically free.
Live traffic / occupancy	Optional licensed data modules.	Potential recurring provider fees; not required to navigate or view saved parking rates.
AI	Templates/local OCR by default; cloud AI optional.	Optional inference costs; local models also consume device resources.
Distribution / support	Protected production signing and reproducible releases.	Store fees where applicable, support, testing devices, legal review and development labor.

Download cost rather than trip cost

Budget formula: download bandwidth = active devices x downloaded package size x number of full-equivalent downloads. For example, 100 devices downloading an illustrative 500 MB package once is about 50 GB; 1,000 devices is about 500 GB, before retries/updates. 500 MB is an example, not a measured Cebu package size. Measure actual map/graph/search size before committing to storage requirements or a hosting plan.

Each additional ride on an already downloaded region need not make any map/navigation cloud request. Updates can be optional Wi-Fi downloads with clear package age. Separate small parking updates from larger map/graph releases. Prefer a budget-capped distribution service; free tiers are conveniences, not long-term guarantees.

Planning allowance

Keep a provisional $0-30/month small-beta hosting allowance only as an internal estimate; a quote needs measured package size, riders, downloads and bandwidth rates. Offline routing can require more implementation work than the previous commercial-SDK plan, so do not treat its earlier labor estimate as a fixed quote. Google Play registration is $25 once and applicable digital subscriptions may incur store fees. [S37,S38]

Commercial pricing remains a fallback comparison

Google Navigation bills destinations rather than map-display minutes; existing-destination reroutes are not separately charged under its documented SDK billing. Mapbox bills navigation sessions/users. Saving data locally does not automatically remove SDK charges or grant offline content rights. Neither SDK is in the offline baseline. [S48,S49]

Can you market it in ATR 160 groups?

Potentially, but this research cannot establish unconditional permission or QJMotor approval. An independent compatibility app and a misleading official product are different propositions. No ATR-160-specific public third-party projection approval/prohibition was located on the official Philippine site. That absence is not authorization. The actual owner manual, dealer warranty, dashboard firmware terms and accepted Carbit agreement still need review. [S34]

Before public beta advertising

Use your own name and artwork. Describe the tested model/phone/firmware accurately and avoid official/authorized/certified claims, QJMotor logos, copied Carbit artwork or unsupported warranty promises. Ask the specific group administrator about promotional rules; membership alone does not settle posting permission. Do not advertise all ATR 160 units as supported from a single successful test.

Before charging or broad distribution

Obtain a Philippine IP/software review of protocol research, code provenance, contractual restrictions and marketing language. Section 185 of the Philippine IP Code recognizes that interoperability decompilation may constitute fair use under its criteria; it is not a blanket commercial exemption or a substitute for analyzing contracts and other rights. Do not ship proprietary SDK binaries, extracted keys/assets or activation bypasses. Review GPL/AGPL obligations before incorporating reference code. [S35]

For any optional commercial map data, clear terms for commercial navigation and projection. For the baseline, audit open-source component obligations, OSM attribution/ODbL and third-party data rights. The open-source decision does not establish QJMotor/Carbit approval. Google specifically restricts embedded vehicle use in general terms, while its SDK policies address permitted mobile navigation. Do not assume Android Auto examples authorize EasyConn/PXC. TomTom and HERE also need account-specific confirmation. [S6-S8]

Treat identifiable location/history and diagnostic uploads as personal information. Document purpose, lawful basis, recipients, retention, security and deletion. Keep trip sharing and telemetry optional, minimize collection and publish a truthful privacy notice. The Philippine Data Privacy Act requires transparency, legitimate purpose and proportionality. [S36]

Review business/tax registration, consumer terms, refunds, digital purchase rules and support obligations for the actual launch structure. Replace the prototype signing setup with a protected production key and finalized application identity; the shared test key is unsuitable for commercial release. Complete Play policies and testing, including the applicable new-personal-account closed test requirement. [S37-S39]

Reviewable questions to send yourself - no messages sent

QJMotor/distributor: Does an independent Android navigation app connecting through the existing dashboard projection interface, with no firmware modification or engine access, conflict with the Philippine ATR 160 warranty or software terms? Please identify applicable terms and permitted compatibility wording.

Map provider: May our commercial Android app render maps and guidance on the phone and encode that scene as H.264 to an existing motorcycle TFT over Wi-Fi Direct/PXC, with the phone screen off? Which license, attribution, storage limits and billing units apply? Does the exact API/SDK support Philippine 160 cc routes and restrictions?

Development sequence and launch evidence

Stage	Deliverable	Exit condition
1. Local engine spike	Cebu map plus on-device route/search; dedicated dashboard rendering.	Networking disabled: calculate route, miss a turn, reroute, voice and screen-off rendering. Measure graph size, memory and battery.
2. Navigation + arrival MVP	QR/saved bike, Fit my TFT calibration, adaptive layout, navigation, offline parking pins with inline known rates and no parking footer or TFT selection.	10-20 verified sample lots; correct motorcycle tariffs/entrances; arrival UI readable before destination.
3. Reliability beta	Atomic package updates, reconnect, stale GPS, no-data states and production signing.	Long sessions and interruption tests across selected phones/firmware; no mandatory cloud dependency.
4. Rider beta	Small opt-in Cebu ATR testing with correction workflow.	Route, tariff and usability issues resolved; privacy, support and compatibility matrix ready.
5. Paid launch	Provenance/licenses, final branding, offline packages, store distribution.	Cost caps, legal review and truthful marketing; recurring data promises match budget.
6. Optional enhancements	Operator partnerships, traffic, occupancy feeds and optional AI authoring/planning.	Licensed coverage, explicit costs and complete offline fallback; no guessed rates or space availability.

Engineering preparation

Migrate packaging to a standard Android Gradle project with version catalog, reproducible builds, CI compile/lint, protocol regression tests and recorded non-sensitive fixtures. Isolate display and navigation from Activity lifecycle. Keep credentials out of the repo, bind dashboard traffic to the correct interface, and leave Internet traffic on an available Internet network. Validate current Android foreground location/connected-device rules and store target API at release time.

Product preparation

Create the selected phone and bike design as editable production UI; specify day/night colors, type sizes, route symbols, state/error messages and attribution slots. Produce a clean standalone logo and app icon after selecting the concept. Generated maps, timings and routes are illustrations, not geographic assets or a functioning navigation build.

What remains unresolved

On-device engine integration and measured Cebu package size; 160 cc access correctness; screen-off rendering; long-ride reliability; verified parking rate/entrance coverage and maintenance; actual ATR warranty/software terms; name/trademark clearance. Optional traffic/occupancy/AI contracts are not baseline blockers. These are concrete development/launch gates, not reasons to keep expanding the diagnostic.

Research method

Official documentation, vendor pricing/terms, primary GitHub repositories/releases and Maven metadata were reviewed. Mobbin supplied a route-planning UX reference. Product Design and built-in image generation produced three visual/branding concepts. No competitor code or assets were imported into the app during this research. Websites and metadata can change; re-check terms, published artifacts and prices at integration and release.

Sources 1-15

Checked 19 October 2026. Click a URL to open the source. Repository release/license snapshots are not endorsements or support guarantees.

S1 - TomTom pricing: current monthly API allowances and SDK sales terms
https://docs.tomtom.com/pricing

S2 - TomTom legacy routing: motorcycle beta and restriction limitations
https://docs.tomtom.com/routing-api/documentation/tomtom-maps/v1/calculate-route

S3 - TomTom Orbis v3 migration: car and taxi travel modes
https://docs.tomtom.com/routing-api/documentation/tomtom-orbis-maps/v3/product-information/migration-guide

S4 - TomTom Android SDK 2.6 release notes
https://docs.tomtom.com/navigation/android/releases/versions/2.6

S5 - TomTom traffic country coverage
https://docs.tomtom.com/traffic-api/documentation/tomtom-maps/v1/product-information/market-coverage

S6 - Google Navigation SDK policies
https://developers.google.com/maps/documentation/navigation/android-sdk/policies

S7 - Google Maps Platform terms, section 3.2.3
https://cloud.google.com/maps-platform/terms

S8 - Google service-specific terms: Navigation, Places and Places UI Kit
https://cloud.google.com/maps-platform/terms/maps-service-terms

S9 - Google pricing: Navigation Request SKU
https://developers.google.com/maps/billing-and-pricing/pricing

S10 - Google supported two-wheel countries, including Philippines
https://developers.google.com/maps/documentation/routes/coverage-two-wheeled

S11 - HERE SDK Android routing and offline capabilities
https://docs.here.com/here-sdk/docs/android-routing

S12 - HERE scooter routing options
https://docs.here.com/here-sdk/docs/android-routing-options

S13 - HERE Android current guide version
https://docs.here.com/here-sdk/docs/android-readme

S14 - Mapbox pricing: Navigation SDK v3 metered trips
https://www.mapbox.com/pricing

S15 - Mapbox Directions: supported profiles
https://docs.mapbox.com/api/navigation/directions/

Sources 16-30

Checked 19 October 2026. Click a URL to open the source. Repository release/license snapshots are not endorsements or support guarantees.

S16 - Waze Transport SDK limitations and 1 November 2026 support end
https://developers.google.com/waze/intro-transport

S17 - MapLibre Android 13.7.1 release
https://github.com/maplibre/maplibre-native/releases/tag/android-v13.7.1

S18 - MapLibre Navigation Android: README and MIT license
https://github.com/maplibre/maplibre-navigation-android

S19 - MapLibre Navigation release 5.0.0
https://github.com/maplibre/maplibre-navigation-android/releases/tag/5.0.0

S20 - Kotlin release table
https://kotlinlang.org/docs/releases.html

S21 - Compose BOM official guide
https://developer.android.com/develop/ui/compose/bom

S22 - CameraX stable releases
https://developer.android.com/jetpack/androidx/releases/camera

S23 - MOTO-HUB open-source CORE and closed-source ADV-SOLO distinction
https://github.com/vincenzobpt/MOTO-HUB

S24 - ridedaemon transport reference
https://github.com/charliecharlieO-o/ridedaemon-lib

S25 - open-cflink repository: no top-level license detected
https://github.com/ionutradu252/open-cflink

S26 - Valhalla routing engine and MIT license
https://github.com/valhalla/valhalla

S27 - GraphHopper routing engine and Apache-2.0 license
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S28 - Organic Maps offline navigation reference
https://github.com/organicmaps/organicmaps

S29 - Android location and battery scenarios
https://developer.android.com/develop/sensors-and-location/location/battery/scenarios

S30 - Android accessibility: contrast and 48 dp targets
https://developer.android.com/guide/topics/ui/accessibility/apps

Sources 31-45

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S31 - Google Design for Driving principles
https://developers.google.com/cars/design/design-foundations

S32 - OSM public tile usage policy
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S33 - Public Nominatim usage policy
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S34 - QJMotor Philippines official site and distributor contact
https://www.qjmotorph.com/

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https://wipolex-res.wipo.int/edocs/lexdocs/laws/en/ph/ph198en.html

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S37 - Google Play developer registration fee
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S39 - Google Play testing requirements for new personal accounts
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S40 - Mobbin route preview reference reviewed
https://mobbin.com/screens/9fae97f1-4455-4041-bb5a-d8af97e4e4b0

S41 - ZXing latest release metadata
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S42 - OkHttp published version metadata
https://repo.maven.apache.org/maven2/com/squareup/okhttp3/okhttp/maven-metadata.xml

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S44 - OSM parking and entrance data
https://wiki.openstreetmap.org/wiki/Tag:amenity%3Dparking

S45 - OSM price/charge tag
https://wiki.openstreetmap.org/wiki/Key:charge

Sources 46-49

Checked 19 October 2026. Click a URL to open the source. Repository release/license snapshots are not endorsements or support guarantees.

S46 - OSM fee tag; missing values are not verified free parking
https://wiki.openstreetmap.org/wiki/Key:fee

S47 - OSM dedicated motorcycle parking
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S48 - Google Navigation actual billing triggers
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https://docs.mapbox.com/android/navigation/guides/pricing/